

ANSWER KEY — Integration Lesson 7 Test: The Fundamental Theorem of Calculus

For the grader · full working shown · give credit for correct method with small arithmetic slips · deduct if $F(a)$ is not subtracted

1. Use the Fundamental Theorem to evaluate $\int_0^4 5 \, dx$, then check with a rectangle.

Work: $F(x) = 5x$ (derivative of $5x$ is 5 ✓). $F(4) = 20$, $F(0) = 0$, so $20 - 0 = 20$. Rectangle check: width $4 \times$ height $5 = 20$ ✓. **Answer: 20.**

2. Use the Fundamental Theorem to evaluate $\int_0^3 2x \, dx$, then check with a triangle.

Work: $F(x) = x^2$ (derivative is $2x$ ✓). $F(3) = 9$, $F(0) = 0$, so $9 - 0 = 9$. Triangle check: base 3 , height $2 \times 3 = 6$, area $\frac{1}{2} \times 3 \times 6 = 9$ ✓. **Answer: 9.**

3. Evaluate $\int_0^3 x^2 \, dx$.

Work: $F(x) = x^3/3$ (derivative: $3x^2/3 = x^2$ ✓). $F(3) = 27/3 = 9$, $F(0) = 0$, so $9 - 0 = 9$. **Answer: 9.**

4. Evaluate $\int_1^2 x^3 \, dx$. Give the exact fraction and a decimal.

Work: $F(x) = x^4/4$ (derivative: $4x^3/4 = x^3$ ✓). $F(2) = 16/4 = 4$, $F(1) = 1/4$. Then $4 - 1/4 = 16/4 - 1/4 = 15/4$. **Answer: $15/4 = 3.75$.** (Watch for the classic error of forgetting $F(1) = 1/4$ and answering 4 .)

5. Evaluate $\int_0^2 (3x^2 + 4) \, dx$.

Work: term by term: $3x^2 \rightarrow x^3$, $4 \rightarrow 4x$, so $F(x) = x^3 + 4x$ (derivative: $3x^2 + 4$ ✓). $F(2) = 8 + 8 = 16$, $F(0) = 0$. **Answer: 16.**

6. Evaluate $\int_1^5 (2x + 1) \, dx$ with the FTC, then verify with the trapezoid method.

Work: $F(x) = x^2 + x$ (derivative: $2x + 1$ ✓). $F(5) = 25 + 5 = 30$, $F(1) = 1 + 1 = 2$, so $30 - 2 = 28$. Trapezoid check: heights $2(1) + 1 = 3$ and $2(5) + 1 = 11$; width $5 - 1 = 4$; area $= \frac{1}{2}(3 + 11) \times 4 = 7 \times 4 = 28$ ✓. **Answer: 28.**

7. A student evaluates $\int_1^3 2x \, dx$ as "x² at 3 is 9, answer 9." Find and fix the mistake.

Work: the antiderivative and $F(3) = 9$ are correct, but the student never subtracted $F(1)$ — the FTC is $F(b) - F(a)$, a difference of TWO readings (like an odometer: end reading minus start reading). $F(1) = 1$, so the correct value is $9 - 1 = 8$. **Answer: the mistake is forgetting to subtract $F(1)$; correct answer 8.**

8. Show with one line of algebra why the $+C$ is harmless in the FTC.

Work: $(F(b) + C) - (F(a) + C) = F(b) - F(a) + C - C = F(b) - F(a)$. The C is added once and subtracted once, so it cancels for every possible value — all antiderivatives in the family give the same definite integral, and we may use the $C = 0$ one. **Answer: the C 's cancel in the subtraction, leaving $F(b) - F(a)$.**

9. A pump's flow rate is $r(t) = 8t$ liters per hour. Find the total water pumped from $t = 1$ to $t = 2$ hours.

Work: total $= \int_1^2 8t \, dt$. $F(t) = 4t^2$ (derivative: $8t$ ✓). $F(2) = 16$, $F(1) = 4$, so $16 - 4 = 12$. Units: (liters/hour) $\times$ hours = liters. **Answer: 12 liters.**

10. Challenge: evaluate $\int_0^3 (x^2 - x + 2) \, dx$.

Work: term by term: $x^2 \rightarrow x^3/3$, $-x \rightarrow -x^2/2$, $2 \rightarrow 2x$, so $F(x) = x^3/3 - x^2/2 + 2x$ (derivative: $x^2 - x + 2$ ✓). $F(3) = 27/3 - 9/2 + 6 = 9 - 4.5 + 6 = 10.5$. $F(0) = 0$. So $10.5 - 0 = 10.5$. **Answer: $21/2 = 10.5$.**